OPEN Research Article

Sex, Racial, and Ethnic Diversity of US Musculoskeletal Oncology Fellows, a 10-year Analysis

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ABSTRACT

Background: Diversity in the orthopaedic surgery workforce is essential for addressing healthcare disparities, enhancing patient-physician trust, and improving outcomes. Despite a growing diverse US population, orthopaedics remains one of the least diverse surgical specialties. However, diversity within musculoskeletal oncology fellowships has not been examined. Identifying disparities in representation is a critical step toward fostering a more inclusive workforce that better reflects the patient population. This study asked (1) is female representation equitable in musculoskeletal oncology fellowships, and (2) are individuals from underrepresented racial and ethnic backgrounds equally represented?

Methods: We analyzed publicly available demographic data for US medical graduates, orthopaedic surgery residents, and musculoskeletal oncology fellows from 2013 to 2022. Sex and race/ethnicity were extracted, and annual percentages for each demographic category were assessed. Participation-to-prevalence ratios (PPRs) were calculated to determine overrepresentation (>1.2), equitable representation (0.8 to 1.2), and underrepresentation (<0.8) relative to US Census data.

Results: Female fellows were underrepresented (PPR = 0.73). Hispanic and Black fellows were also underrepresented, with PPRs of 0.21 and 0.29, respectively. Notably, there were no Native Hawaiian/Pacific Islander or Native American/Alaskan Native fellows in this subspecialty. Asian fellows were overrepresented (PPR = 1.47), and White fellows were equitably represented (PPR = 1.08).

Conclusion: Musculoskeletal oncology fellowships continue to demonstrate sex and racial/ethnic disparities, with notable underrepresentation of female, Hispanic, Black, Native Hawaiian/Pacific Islander, and Native American/Alaskan Native fellows. Despite increasing diversity at earlier stages of medical training, these improvements are not reflected in fellowship representation, highlighting the need for targeted efforts to promote inclusivity.

In the United States, there has been a substantial change in demographics especially among racial, ethnic, and gender minorities as of late.[1] The increased diversity has proven beneficial in several aspects of society, especially the workforce.[2] This also holds true for the clinical setting. Racial, ethnic, and gender similarities between patients and physicians can strengthen patient trust and communication with their physicians as well as improve their surgical outcomes.[3-5]

Despite these increasing trends in diversity, individuals from underrepresented racial, ethnic, and gender backgrounds in medical school remain far below their population makeup in the US Census.[6] This trend holds true for the orthopaedic surgery workforce because it remains one of the least diverse surgical specialties in the United States.[7-9] This can be attributed to several unique barriers present in the orthopaedic workforce. For instance, increased discrimination in orthopaedic surgery has been found to be directed toward racial and ethnic minority residents and attendings, and women have reported experiencing unconscious biases and minimal mentorship from other orthopaedic surgeons.[10-12]

The number of fellowship positions in orthopaedic subspecialities is increasing over time, including for musculoskeletal oncology.[13,14] However, the demographic trends within musculoskeletal oncology fellowships have yet to be studied and described. This is a field of orthopaedics where mortality is a genuine concern because malignancies of the musculoskeletal system have markedly lower prognoses compared with other forms of cancers.[15,16] Studies have shown racial disparities in the outcomes of primary bone tumors, with Black patients experiencing lower survival rates compared with White patients.[17] For this reason, diversity and patient-physician concordance may be more important for this subspeciality than others. Therefore, it is paramount to compare the trends in demographic composition changes among orthopaedic oncology fellows against trends for orthopaedic residents and medical students to understand discrepancies that may exist. Although disparities exist across all areas of orthopaedic surgery, musculoskeletal oncology represents a uniquely high-stake subspecialty in which inequities may have particularly profound consequences. Patients with primary bone and soft tissue sarcomas often require complex, multidisciplinary care, prolonged follow-up, and shared decision making in the setting of potentially life-threatening disease. Studies have linked race, insurance status, and socioeconomic factors to delayed presentation, limited access to care, and poorer survival in musculoskeletal oncology.[18] In this context, workforce diversity and patient-physician concordance may play an especially important role in trust, communication, and access to timely care, making musculoskeletal oncology a critical subspecialty for focused evaluation. Analyzing diversity at the fellowship level may reveal barriers in recruitment, mentorship, or access that are not visible in broader training data and are critical to building a more representative workforce. Thus, our research questions were as follows: (1) Is female representation equitable in musculoskeletal oncology fellowships? (2) Are individuals from different racial and ethnic backgrounds equitably represented in musculoskeletal oncology fellowships relative to their representation in the musculoskeletal oncology training pipeline?

Methods

This was a longitudinal descriptive study. Institutional review board was not required for this study because publicly available data were used. Demographic data on sex and race/ethnicity of medical students and trainees in the United States from 2013 to 2022 were extracted. We categorized the data from 2013 as corresponding to the 2013 to 2014 academic year and followed this pattern for subsequent years. The American Association of Medical Colleges provided demographic data on medical students who graduated from schools that award a Doctor of Medicine degree.[19] The Accreditation Council for Graduate Medical Education annual data resource books provided demographic data on orthopaedic surgery residents and musculoskeletal oncology fellows.[20] Sex was reported as male or female in the data. We used the term “sex” because the Accreditation Council for Graduate Medical Education uses male/female labels in their reports, although they refer to these labels as “gender.” Race and ethnicity were reported as a combined variable in the data sources and were classified for analysis as Hispanic, Black or African American (“Black”), Asian, White, American Indian or Alaska Native, and Native Hawaiian or Other Pacific Islander. Those with a racial/ethnic categorization of multiple, other, or unknown were not included.

The total number of medical students, orthopaedic residents, and musculoskeletal oncology fellows per year was extracted as well as the annual number for each demographic category. The percentages were calculated for each group to reflect demographic representation over time.

Participation-to-prevalence ratios (PPRs) were calculated and used to classify demographic group representation in the musculoskeletal oncology specialty. PPRs were calculated by taking the percentage of a given demographic group in the musculoskeletal oncology subspecialty and dividing that by the percentage of that given demographic group in the entire US population using the US Census 2020 data.[21] A PPR > 1.2 was deemed to be an overrepresentation of a given demographic, while a PPR < 0.8 was deemed to be an underrepresentation. A PPR between 0.8 and 1.2 was classified as equitably represented. This statistical methodology has been previously described in several studies.[22-25]

Results

Female Representation Among Musculoskeletal Oncology Trainees

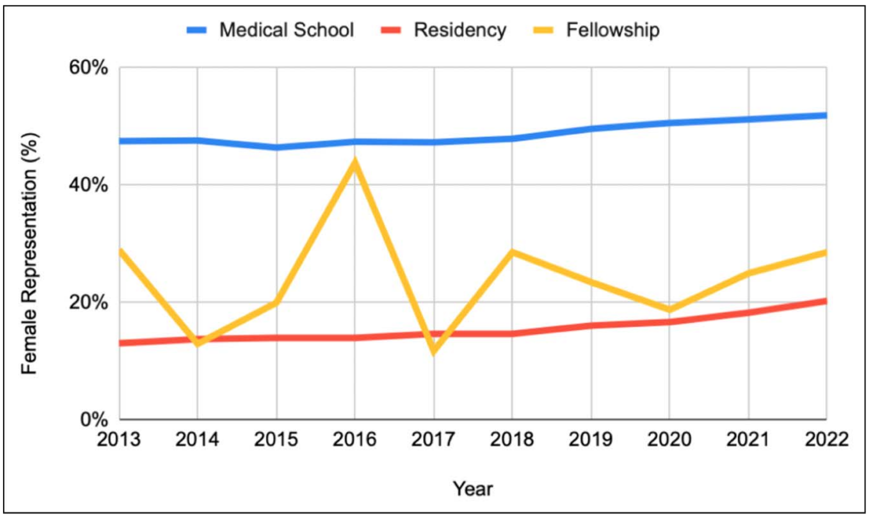
The representation of women showed an overall increase at the medical school and residency stages (Figure 1). From 2013 to 2022, the proportion of female medical school graduates increased from 47.5% to 51.9%. During this same period, female representation among orthopaedic surgery residents grew from 13.1% to 20.3%. However, the proportion of women pursuing musculoskeletal oncology fellowships saw minimal change, decreasing from 29.0% to 28.6% from 2013 to 2022. Overall, across the study period, female representation was highest among medical school graduates (48.7%), followed by musculoskeletal oncology fellows (24.2%), with the lowest representation among orthopaedic surgery residents (15.6%). Female fellows were notably underrepresented relative to the US population (PPR = 0.73), while male fellows were overrepresented (PPR = 1.27).

Racial and Ethnic Representation in Musculoskeletal Oncology Trainees

From 2013 to 2022, the proportion of Hispanic trainees among medical school graduates showed a small increase from 5.1% to 5.7% (Figure 2). Hispanic representation among orthopaedic surgery residents presented a more dramatic increase from 2.8% to 7.4% during the same period. However, the representation of Hispanic trainees aspiring to fellowships in musculoskeletal oncology showed a substantial decrease from 2013 to 2022, decreasing from 21.4% to 7.1%. Overall, Hispanic individuals were most frequently represented among medical school graduates (5.4%) compared with orthopaedic surgery residents (4.2%) and musculoskeletal oncology fellows (4.7%) throughout the study period.

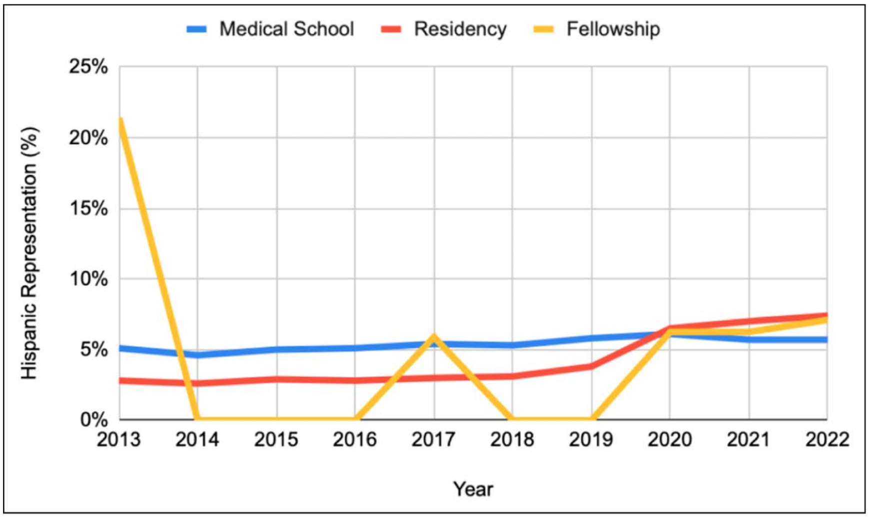
In recent years, the representation of Black medical school graduates showed a slight increase from 5.8% to 6.7% from 2013 to 2022. Black representation among orthopaedic surgery residents also demonstrated an increase from 3.4% to 5.2% within the same period (Figure 3). Among Black trainees pursuing fellowships in musculoskeletal oncology, representation was 0% in both

Figure 1



Graph showing female trainees in the emerging orthopaedic musculoskeletal oncology workforce compared with total trainees.

Figure 2



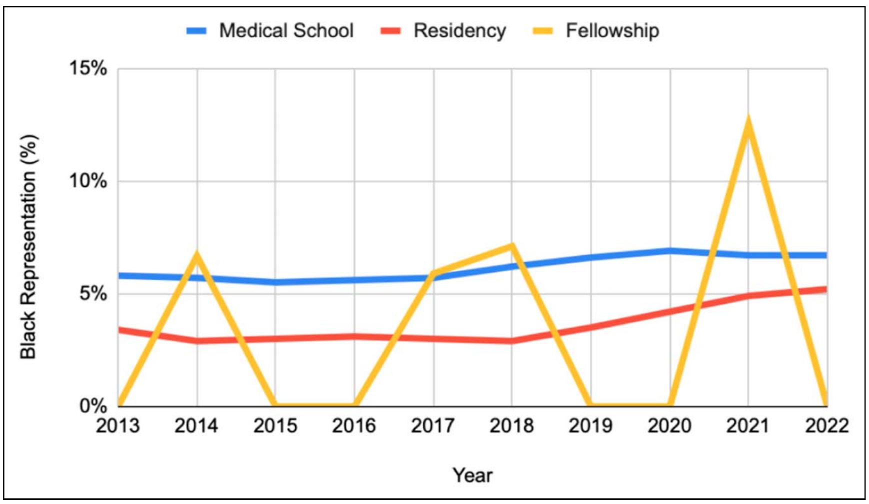
Graph showing Hispanic trainees in the emerging orthopaedic musculoskeletal oncology workforce compared with total trainees.

2013 and 2022, although fluctuations were seen during the period. Overall, the average representation of black trainees among medical school graduates in recent years was 6.1%, followed by orthopaedic surgery residents at 3.6% and musculoskeletal oncology fellows at 3.2%.

Across the study period, the representation of Asian trainees increased among medical school graduates from 20.5% to 23.5%. Similarly, the percentage of Asian orthopaedic surgery residents increased from 8.9% to 13.3%. Asian representation among musculoskeletal oncology fellows showed no overall trend from 2013 to 2022 despite fluctuations, remaining at 21.4% (Figure 4). Overall during the study period, there was the greatest Asian representation among medical school graduates (21.6%), followed by musculoskeletal oncology fellows (14.5%) and orthopaedic surgery residents (10.2%).

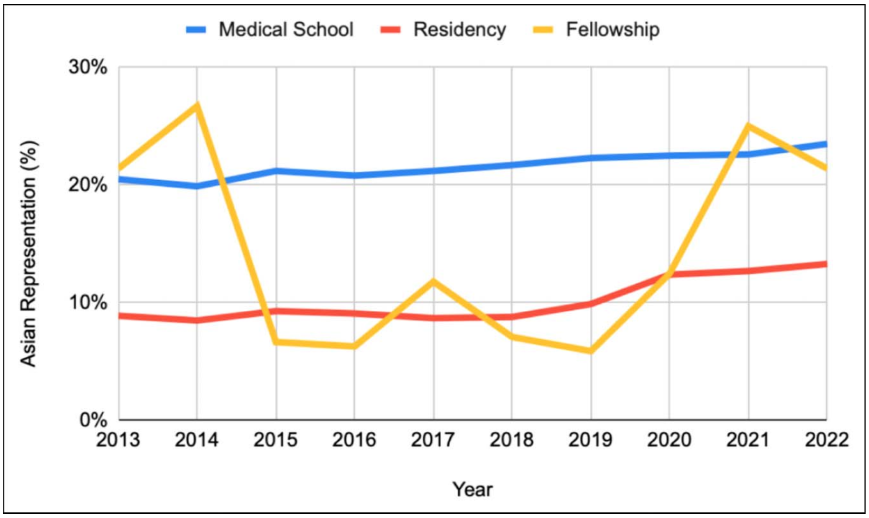
The representation of White trainees among medical school graduates showed a decrease from 57.8% to 49.4% from 2013 to 2022. However, the percentage of White orthopaedic surgery residents increased from 64.7% to 67.7% across the same period. In addition, White representation among musculoskeletal oncology trainees showed an increase from 57.1% to 64.3% from 2013 to 2022 (Figure 5). Across the study period, the greatest representation of White trainees was seen among orthopaedic surgery residents (64%), followed by musculoskeletal oncology fellows (63.2%) and medical school graduates (54.6%).

Figure 3



Graph showing Black trainees in the emerging orthopaedic musculoskeletal oncology workforce compared with total trainees.

Figure 4



Graph showing Asian trainees in the emerging orthopaedic musculoskeletal oncology workforce compared with total trainees.

Regarding racial and ethnic groups, Asian fellows (PPR = 1.47) were overrepresented, whereas Black (PPR = 0.29) and Hispanic (PPR = 0.21) fellows were underrepresented. White fellows (PPR = 1.08) were equitably represented. In addition, over the study period, there were no Native Hawaiian/Pacific Islander fellows or Native American/Alaskan Native fellows in this field.

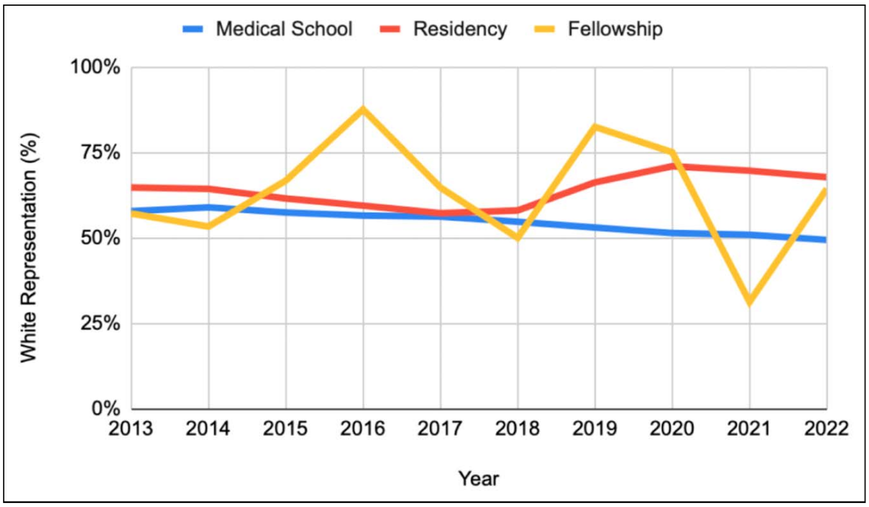
Discussion

Orthopaedic surgery is considered one of the least diverse surgical specialties. Workforce diversity may mitigate sex and racial/ethnic disparities in outcomes for patients. We found an underrepresentation of female fellows as well as Hispanic, Black, Native Hawaiian/Pacific Islander, and Native American/Alaskan Naive fellows, while overrepresentation was observed for male and Asian fellows. These results highlight the need to increase diversity within the musculoskeletal oncology field.

Female Representation Among Musculoskeletal Oncology Trainees

Female underrepresentation in the orthopaedic field has been well documented in the literature.[8,26] Although our findings revealed a steady increase in female representation at the medical school and residency levels, this

Figure 5



Graph showing White trainees in the emerging orthopaedic musculoskeletal oncology workforce compared with total trainees.

trend did not extend to musculoskeletal oncology fellowships, where female participation has remained stagnant. Similar underrepresentation has been observed across other orthopaedic subspecialties such as sports, adult reconstruction, foot and ankle, hand, trauma, and spine.[27-32] Several factors contribute to the persistent gender disparity in orthopaedics. Gender bias and discrimination are substantial obstacles because women often face stereotypes questioning their physical strength and competence in a male-dominated field.[33,34] Importantly, musculoskeletal oncology fellowship directors are mostly White (84%) and male (79%).[35] Despite these barriers, studies have shown that female orthopaedic surgeons achieve outcomes comparable with their male counterparts.[36] In addition, lack of mentorship and limited early exposure to orthopaedics during medical training discourages many women from aspiring to this specialty.[34] Without mentors to provide guidance and support, female medical students may struggle to navigate the challenges of the field. Compounding these issues are workplace dynamics, where women report higher incidences of sexual harassment and gender bias, creating a hostile environment that can deter them from entering or remaining in orthopaedics.[33,37] The scarcity of female role models in leadership positions further hampers efforts to attract and retain women in the field. Although individual preferences undoubtedly play a role in specialty choice, these preferences may be shaped by broader systemic factors, including mentorship access, early exposure, and perceptions of bias or belonging, which influence whether women, and certain ethnic groups, view musculoskeletal oncology as an accessible or welcoming career path.[38] Organizations such as the Ruth Jackson Orthopaedic Society (RJOS) play a crucial role in addressing these barriers by providing scholarships, mentorship opportunities, and advocacy programs.[39] Notably, scholarships from RJOS have demonstrated a measurable effect, with 80% of recipients aspiring to careers in orthopaedics compared with 44.9% of nonrecipients.[40] By fostering mentorship connections and promoting diversity through education and advocacy, RJOS helps create a more inclusive environment for women in orthopaedics. To build a more equitable future, particularly within musculoskeletal oncology, it is vital to address these systemic barriers. Efforts to dismantle gender bias, enhance mentorship opportunities, and support women in orthopaedics are essential to fostering a culture that enables women to envision and achieve successful careers in this field.

Racial and Ethnic Representation in Musculoskeletal Oncology

Racial disparities in outcomes after orthopaedic surgeries have been well documented in the literature.[41] Specifically to musculoskeletal oncology, Elsamadicy et al[42] showed that non-White patients had a lower survival rate from primary osseous neoplasms of the spine. Furthermore, disparities in clinical care have been documented, with White patients being more likely to undergo surgery for osseous spinal neoplasms compared with patients from other racial groups.[43] Another study in 2024 showed that Black patients had lower survival from a high-grade soft tissue sarcoma compared with White patients.[44] A recent systematic review found 25 studies documenting how social determinants of health (including race/ethnicity) are associated with worse outcomes with primary bone tumors.[17] Another systematic review identified disparities in musculoskeletal oncology outcomes linked to socioeconomic factors, race/ethnicity, and insurance status.[18] Racial and ethnic diversity in orthopaedics is essential for building patient trust, enhancing communication, and addressing racial disparities in health care. Despite this critical need, orthopaedics remains one of the least diverse medical specialties.[7] Our study revealed that Asian trainees are overrepresented in musculoskeletal oncology, whereas Black, Hispanic, Native Hawaiian/Pacific Islander, and Native American/Alaskan Native trainees are underrepresented. These findings align with trends observed across various orthopaedic fellowship programs such as sports, adult reconstruction, foot and ankle, hand, trauma, and spine.[27-32] For example, in a recent study on orthopaedic sports medicine fellowships, the representation of female, Hispanic, Black, and Asian trainees was shown to decrease at each stage of the training pipeline, while the representation of White trainees increased.[27] Interestingly, White trainees are equitably represented in musculoskeletal oncology, a result that contrasts with other orthopaedic subspecialties where White trainees are often overrepresented. Although this suggests that musculoskeletal oncology may be comparatively more inclusive, meaningful gaps persist for many underrepresented racial and ethnic groups.

Several factors contribute to the lack of racial and ethnic diversity within orthopaedics. Implicit bias in the selection process for training programs and experiences of racial microaggressions are substantial barriers. For instance, Brooks et al[10] reported that most Black orthopaedic surgeons experienced racial microaggressions during their residency training, which can discourage minority candidates from pursuing or remaining in the field. In addition, the absence of diverse role models and mentors further perpetuates underrepresentation. Rama et al. highlighted that a lack of visible minority mentors in orthopaedics can deter minority medical students from considering the specialty as a viable career path.[37] In addition, underrepresented minority trainees also face higher attrition rates, which further diminishes their representation in fellowship programs.[45] Addressing these disparities will require a multifaceted approach. Early exposure to musculoskeletal oncology during medical school and residency can broaden awareness and interest among underrepresented groups. Structured mentorship programs, particularly those that connect students with diverse role models, can support retention and career development. In fact, several organizations are working to address these challenges and promote diversity in orthopaedics. The J. Robert Gladden Orthopaedic Society focuses on mentorship, advocacy, and research to support underrepresented trainees and reduce disparities in musculoskeletal care.[46] The American Association of Latino Orthopaedic Surgeons advances the careers of Latino orthopaedic surgeons through education, mentorship, and social justice initiatives while inspiring Latino medical students to enter the field.[47] Similarly, Nth Dimensions enhances diversity in orthopaedic surgery by providing women and underrepresented minorities with early exposure, hands-on experience, and mentorship through its Orthopaedic Summer Internship Program.[48] Institutions and fellowship programs should also adopt more transparent and inclusive recruitment practices, track diversity metrics, and foster environments that promote belonging and equity. Recent findings in orthopaedic sports medicine show that programs with visible diversity, equity, and inclusion (DEI) statements and designated DEI roles are associated with higher representation of women among both faculty and fellows. Such initiatives may positively shape program culture, signal inclusivity, and expand mentorship opportunities.[49] Specialty-specific efforts within musculoskeletal oncology that prioritize DEI may have similar benefits. Addressing these disparities will require targeted, multilevel interventions. Early exposure to musculoskeletal oncology during medical school and residency, particularly through structured rotations, research opportunities, and mentorship, may increase interest among underrepresented trainees. At the fellowship level, programs can adopt more transparent and inclusive recruitment practices, including holistic application review, intentional outreach to diverse applicants, and increased representation among faculty and leadership. Institutions may also benefit from tracking diversity metrics, supporting formal mentorship programs, and designating leadership roles focused on diversity, equity, and inclusion. Addressing these systemic issues is vital to fostering a more inclusive and diverse orthopaedic workforce. By doing so, the specialty can improve patient care, build trust among diverse populations, and help reduce racial disparities in musculoskeletal health outcomes. We acknowledge that factors beyond structural barriers may also influence fellowship selection. Musculoskeletal oncology may be perceived by some trainees as less attractive due to its limited number of positions, emotionally demanding patient population, or concerns regarding work-life balance.

Limitations

This study has several limitations due to the integration of data from multiple sources. Variations in the classification and availability of racial and ethnic categories over the study period and across the data sources may have influenced our ability to conduct accurate temporal and comparative analyses of demographic representation. For instance, we excluded categories such as "other," "multiracial," and "unknown" from our analyses to ensure consistent comparisons across data sets. Although self-reported data are considered the standard in sociodemographic research, current data sets from the American Medical Association and the Association of American Medical Colleges do not capture the full complexity of sex/gender, race, and ethnicity. Another limitation of this study is the inability to clearly differentiate between sex and gender, which can affect the interpretation of our data. In addition, the available datasets lacked information on the intersectionality of demographic variables, further limiting the scope of our analysis. The focus of this article was restricted to race/ethnicity and sex diversity, leaving out other dimensions of diversity, such as sexual orientation, religious affiliation, and socioeconomic background. Another limitation is that our analysis was restricted to medical school graduates with a Doctor of Medicine degree, excluding Doctors of Osteopathic Medicine, who represent a growing proportion of residents entering orthopaedic surgery. Finally, as the study centered on the US musculoskeletal oncology workforce, its applicability to the global workforce in this specialty remains uncertain.

Conclusion

Our study has revealed substantial underrepresentation of Black, Hispanic, and female trainees in musculoskeletal oncology fellowships. These disparities highlight the need for efforts to increase diversity within the field. Addressing racial/ethnic and gender bias, expanding mentorship opportunities, and creating a more inclusive environment are essential steps to fostering a workforce that better reflects the diverse patient population, ultimately improving care outcomes and reducing healthcare disparities.

References

1. Alba R, Maggio C: Demographic change and assimilation in the early 21st-century United States. *Proc Natl Acad Sci U S A* 2022;119: e2118678119.

2. Rosenkranz KM, Arora TK, Termuhlen PM, et al: Diversity, equity and inclusion in medicine: Why it matters and how do we achieve it?. *J Surg Educ* 2021;78:1058-1065.

3. Takeshita J, Wang S, Loren AW, et al: Association of racial/ethnic and gender concordance between patients and physicians with patient experience ratings. *JAMA Netw Open* 2020;3:e2024583.

4. Wright MA, Murthi AM, Aleem A, Zmistowski B: Patient disparities and provider diversity in orthopaedic surgery: A complex relationship. *J Am Acad Orthop Surg* 2023;31:132-139.

5. Wallis CJD, Jerath A, Coburn N, et al: Association of surgeon-patient sex concordance with postoperative outcomes. *JAMA Surg* 2022;157:146.

6. Morris DB, Gruppuso PA, McGee HA, Murillo AL, Grover A, Adashi EY: Diversity of the national medical student body—Four decades of inequities. *New Engl J Med* 2021;384:1661-1668.

7. Poon S, Kiridly D, Mutawakkil M, et al: Current trends in sex, race, and ethnic diversity in orthopaedic surgery residency. *J Am Acad Orthop Surg* 2019;27:e725-e733.

8. Chambers CC, Ihnow SB, Monroe EJ, Suleiman LI: Women in orthopaedic surgery: Population trends in trainees and practicing surgeons. *J Bone Joint Surg Am* 2018;100:e116.

9. Hoyt BW, Anderson AB, Dingle ME, et al: Racial and gender diversity of physicians accepted to American military orthopaedic and surgical residencies: An 18-year analysis. *JB JS Open Access* 2023;8:e22.00091.

10. Brooks JT, Porter SE, Middleton KK, Carson EW, Ode GE: The majority of Black orthopaedic surgeons report experiencing racial microaggressions during their residency training. *Clin Orthop Relat Res* 2023;481:675-686.

11. Rodarte P, Kammire MS, Israel H, Poon SC, Cannada LK: The other side of conflict: Examining the challenges of female orthopaedic surgeons in the workplace. *Am J Surg* 2023;225:46-52.

12. Ode GE, Brooks JT, Middleton KK, Carson EW, Porter SE: Perception of racial and intersectional discrimination in the workplace is high among Black orthopaedic surgeons: Results of a survey of 274 Black orthopaedic surgeons in practice. *J Am Acad Orthop Surg* 2022;30:7-18.

13. Ruddell JH, Eltorai AEM, DePasse JM, et al: Trends in the orthopaedic surgery subspecialty fellowship match: Assessment of 2010 to 2017 applicant and program data. *J Bone Joint Surg Am* 2018;100:e139.

14. Silvestre J, Nelson CL, Thompson TL, Kang JD: Trends in ACGME accreditation of orthopedic surgery fellowship training. *Orthopedics* 2024; 47:57-63.

15. Lewis DR, Siembida EJ, Seibel NL, Smith AW, Mariotto AB: Survival outcomes for cancer types with the highest death rates for adolescents and young adults, 1975-2016. *Cancer* 2021;127:4277-4286.

16. Whelan J, McTiernan A, Cooper N, et al: Incidence and survival of malignant bone sarcomas in England 1979-2007. *Int J Cancer* 2012;131: E508-E517.

17. Ebrahimi A, Hosseini S, Shahabi S, Ashkani-Esfahani S, Lozano-Calderon S: The impact of social determinants of health on the prognosis of primary bone tumors: A systematic review. *JBJS Rev* 2024;12.

18. Koons A, Smith E, Stephens JC, McKnight NH, Barr J, Ibe IK: Disparities in musculoskeletal oncology. *Curr Rev Musculoskelet Med* 2024;17:527-537.

19. AAMC: FACTS. AAMC. https://www.aamc.org/data-reports/students-residents/data/facts-applicants-and-matriculants. Accessed January 30, 2025.

20. ACGME. *ACGME Data Resource Book*. ACGME; https://www.acgme.org/about/publications-and-resources/graduate-medical-education-data-resource-book/ (2024, January 30, 2025).

21. Census. *Quick Facts United States*. United States Census Bureau; https://www.census.gov/quickfacts/fact/table/US/POP010220 (2020, January 30, 2025).

22. Chen S, Li J, Shu M: Use of participation to prevalence ratio for evaluating the representation status of women in oncology clinical trials. *JAMA Oncol* 2022;8:479.

23. Poon R, Khanijow K, Umarjee S, et al: Participation of women and sex analyses in late-phase clinical trials of new molecular entity drugs and biologics approved by the FDA in 2007-2009. *J Women's Health* 2002; 2013:22.

24. Scott PE, Unger EF, Jenkins MR, et al: Participation of women in clinical trials supporting FDA approval of cardiovascular drugs. *J Am Coll Cardiol* 2018;71:1960-1969.

25. Jin X, Chandramouli C, Allocco B, Gong E, Lam CSP, Yan LL: Women's participation in cardiovascular clinical trials from 2010 to 2017. *Circulation* 2020;141:540-548.

26. Rodgers BM, Moore ML, Mead-Harvey C, et al: How does orthopaedic surgeon gender representation vary by career stage, regional distribution, and practice size? A large-database medicare study. *Clin Orthop Relat Res* 2023;481:359-366.

27. Silvestre J, Slone HS, Newton WN, Kolade OO, Kelly JD IV. Sex, race, and ethnic diversity of the emerging U.S. orthopaedic sports medicine workforce is limited. *Arthroscopy*. 2025;41:2524–2532.

28. Ajayi PT, Murdock CJ, Destine H, Trenchfield D, Aiyer A, Oni J: Trends in racial, ethnic, and gender diversity in orthopaedic surgery adult reconstruction fellowships from 2007 to 2021. *J Arthroplasty* 2023;38: 2232-2236.

29. Joshi A, Kim A, Hsu N, Aiyer A, Thompson JM: A comparison of demographic diversity between orthopaedic surgery residents and ACGME foot and ankle fellows from 2007 to 2022. *Foot Ankle Orthop* 2024;9:24730114241263056.

30. Shittu A, Murdock CJ, Destine H, et al: Trends in racial, ethnic, and gender diversity among hand surgery fellows from 2007 to 2021. *J Hand Surg* 2025;50:98.e1-98.e8.

31. Silvestre J, Ahn J, Dehghan N, Gitajn IL, Slobogean GP, Harris MB: Analysis of the diversity pipeline for the orthopedic trauma surgeon workforce in the United States. *Injury* 2024;55:111695.

32. Trenchfield D, Murdock CJ, Destine H, Jain A, Lord E, Aiyer A: Trends in Racial, Ethnic, and Gender Diversity in Orthopedic Surgery Spine Fellowships from 2007 to 2021. *Spine (Phila Pa 1976)* 2023;48: E349-E354.

33. Day CS, Lattanza L, Van Heest A, Templeton K, Fore JA, Ode GE: AOA critical issues: Gender justice in academic medicine: What it might look like in orthopaedic surgery. *J Bone Joint Surg* 2023;105:1214-1219.

34. Rohde RS, Wolf JM, Adams JE: Where are the women in orthopaedic surgery?. *Clin Orthop Relat Res* 2016;474:1950-1956.

35. Moore ML, Elahi MA, Doan MK, et al: Investigation of fellowship leadership in orthopaedic musculoskeletal oncology. *J Am Acad Orthop Surg Glob Res Rev* 2022;6:e22.00039.

36. Jolbäck P, Rogmark C, Bedeschi Rego De Mattos C, Chen AF, Nauclér E, Tsikandylakis G: The influence of surgeon sex on adverse events following primary total hip arthroplasty: A register-based study of 11,993 procedures and 200 surgeons in Swedish public hospitals. *J Bone Joint Surg* 2022;104:1327-1333.

37. Rama E, Ekhtiari S, Thevendran G, Green J, Weber K, Khanduja V: Overcoming the barriers to diversity in orthopaedic surgery: A global perspective. *J Bone Joint Surg* 2023;105:1910-1919.

38. Winfrey SR, Parameswaran P, Gerull KM, LaPorte D, Cipriano CA: Effective mentorship of women and underrepresented minorities in orthopaedic surgery: A mixed-methods investigation. *JB JS Open Access* 2022;7:e22.00053.

39. Samora JB, Russo C, LaPorte D: Ruth Jackson Orthopaedic Society: Promoting women in orthopaedics. *J Am Acad Orthop Surg* 2022;30:364-368.

40. Vajapey S, Cannada LK, Samora JB: What proportion of women who received funding to attend a Ruth Jackson Orthopaedic Society meeting pursued a career in orthopaedics?. *Clin Orthop Relat Res* 2019;477:1722-1726.

41. Rudisill SS, Varady NH, Birir A, Goodman SM, Parks ML, Amen TB: Racial and ethnic disparities in total joint arthroplasty care: A contemporary systematic review and meta-analysis. *J Arthroplasty* 2023;38:171-187.e18.

42. Elsamadicy AA, Sayeed S, Sherman JJZ, et al: Racial disparities in the management and outcomes of primary osseous neoplasms of the spine: A SEER analysis. *J Neurooncol* 2024;166:293-301.

43. Deb S, Brewster R, Pendharkar AV, Veeravagu A, Ratliff J, Desai A: Socioeconomic predictors of surgical resection and survival for patients with osseous spinal neoplasms. *Clin Spine Surg* 2019;32:125-131.

44. Braswell AC, Jiminez V, Montgomery KB, McLeod MC, Broman KK, Eulo V: Racial and ethnic variation in presentation and outcomes of high-grade soft tissue sarcoma at a Southeastern United States comprehensive cancer center. *Am J Surg* 2024;227:132-136.

45. Haruno LS, Chen X, Metzger M, et al: Racial and sex disparities in resident attrition in orthopaedic surgery. *JB JS Open Access* 2023;8:e22.00148.

46. Brooks JT, Taylor E, Peterson D, Carson E: The J. Robert Gladden Orthopaedic Society: Past, present, and future. *J Am Acad Orthop Surg* 2022;30:344-349.

47. Escalera C, Rodarte P, Navarro RA, et al: The American Association of Latino orthopaedic surgeons: Our history, achievements, and future goals. *J Bone Joint Surg Am* 2024;106:748-754.

48. Mason BS, Ross W, Ortega G, Chambers MC, Parks ML: Can a strategic pipeline initiative increase the number of women and underrepresented minorities in orthopaedic surgery?. *Clin Orthop Relat Res* 2016;474:1979-1985.

49. Lessiohadi NJ, Pai J, Goodman WB, et al: Diversity, equity, and inclusion initiatives and faculty roles in orthopaedic sports medicine fellowships are associated with increased female representation among faculty and fellows. *JB JS Open Access* 2025;10:e24.00198.